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is drawn towards the vertebral column, by contraction of the muscles of the body-wall. As is drawn up, the elastic ribs are bent so as to bring about a decrease in the size of the body cavity and the air from the lungs is forced out. When the muscles relax, the body-cavity recovers its size and air is drawn in.

Syrinx

The larynx does not take part in the production of voice. The **voice box** lies deep down where the trachea divides into two bronchi, and is known as **syrinx**, a structure characteristic of birds. It consists of a chamber with its walls supported by three or four rings of the trachea and the first ring of each bronchus; its inner lining is raised into folds, the vibrations of which is caused by the movement of air results in the production of sound.

Circulatory system

Pigeon has an efficient circulatory system to meet the metabolic demands of flight, but also plays a significant role in maintaining the body temperature. The circulatory system of pigeon includes the heart and blood vessels. The **heart** of the pigeon is **four chambered** with two auricles and two ventricles. There is no sinus venosus. The two **precaval veins** or **superior venae cavae**, a **post caval vein** or **inferior vena cava** opens into the right auricle; the pulmonary aorta and systemic trunks arise from the right and left ventricles respectively. The right side of the heart is completely separated from the left side of the heart by a septum. The right auricle opens into the right ventricle by the **right auriculo-ventricular** aperture and the left auricle into the left ventricle by the **left auriculo- ventricular** aperture. There are **valves** at these apertures, which allows the blood to flow only in one direction, i.e., from the auricle into the ventricle but not backwards. The right auriculo-ventricular valve consists of a single flap without connecting **chordae tendinae**; the valve on the left side has two flaps connected to the **papillary muscles** by chordae tendinae. The pulmonary aorta arises from the right ventricle and the aortic arch from the left ventricle. The pulmonary veins open into the left auricle. There are three **semilunar valves** at the junction of the pulmonary aorta and the right ventricle. The pulmonary aorta divides into two branches, each entering a lung. Only the right aortic arch is present in birds.

The right auricles of the heart receives venous blood from all parts of the body except the lungs, through the **precaval** and **post caval** veins. The right ventricles pumps venous blood into the lungs through the **pulmonary aorta.** The oxygenated blood from the lungs is returned to the left auricle through the **pulmonary veins**. From the left ventricle a single right aortic arch carries oxygenated blood to the different parts of the body. The right half of the heart receives and discharges only **venous blood** and the left half only **arterial blood**. Thus birds possess a **complete double circulation** which includes the **pulmonary circulation** and **systemic circulation**. (Figure 4.31)

The arterial system

The right aortic arch curves over to the right side giving off at the curve the right and the left **innominate arteries**; each